

Sustainable Petroleum Refining: Biodiesel Production

Senior Design Mini Project I

Group I

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Table of Contents

Project Charter	2
Design Plan	2
Introduction	3
Problem Definition & Significance	3
Market Analysis	5
Process Description	8
Competing Processes	8
<i>Creating fuel using crop waste</i>	8
<i>Using rubber tires</i>	9
<i>Using Soybean waste</i>	10
Chosen Process Detailed Description	11
Appendix	13
Design Basis	13
Block Flow Diagram (BFD)	15
Preliminary Process Flow Diagram	16
Rough Material Balance	16
References	19

Project Charter

The objective of this report is to evaluate biodiesel production within petroleum refining. As a team, we will assess competing processes, design a comprehensive process flow diagram (PFD) and perform a rough material balance. The scope of this project is to conduct a market and problem significance analysis, compare competing production processes, justify the sustainability of biodiesel, develop a block flow diagram (BFD), PFD and a rough material balance, and analyze economic and policy factors for biodiesel, while highlighting advantages and disadvantages.

This report will deliver a typed technical report, Project Charter and Design Plan, Introduction, Process Description, Appendix as well as a Gantt Chart of project timeline and delegated responsibilities. Assumptions we will be making in this report include the plant size of 100,000 metric tons/year of biodiesel, feedstock (waste cooking oil), a conversion of 95% triglyceride to FAME, 90-95% methanol recycled and an operating schedule of 330 days/year.

Design Plan

We have established a structured plan to ensure the timely and efficient completion of this report. The Design Plan outlines our approach to research, analysis and report development, as well as the allocation of responsibilities among team members.

This report will be divided into major tasks, with each team member taking responsibility for one or more areas (note: roles are flexible, and team members will collaborate and provide support across sections as needed). The project schedule and roles are illustrated in the Gantt Chart

below, which highlights start and end dates for each major task. The schedule ensures completion of all deliverables prior to the September 15th deadline.

Task	Assigned To	Start Date	End Date
Project Charter (team roles, scope, deliverables)	All	30-Aug	1-Sep
Design Plan (timeline, responsibilities, Gantt chart)	Lindsey	30-Aug	2-Sep
Introduction – Problem definition & significance	Angelica, Omar	2-Sep	7-Sep
Introduction – Market analysis	Lindsey	2-Sep	7-Sep
Competing Processes Research + BFDs	Rishabh	2-Sep	5-Sep
Chosen Process Detailed Description	Lindsey/Angelica	6-Sep	12-Sep
Design Basis (capacity, feedstock, location)	Rishabh	5-Sep	12-Sep
Block Flow Diagram (BFD)	Angelica	7-Sep	9-Sep
Preliminary Process Flow Diagram (PFD)	Ameen	6-Sep	8-Sep
Rough Material Balance	Omar	6-Sep	9-Sep
Compile & First Draft Assembly	Editor	9-Sep	10-Sep
Review & Edits (figures, tables, bibliography)	All	11-Sep	12-Sep
Final Review + Submission Prep	All	13-Sep	14-Sep
Submit Report	Leader	15-Sep	15-Sep

Introduction

Problem Definition & Significance

The world energy system suffers from drastic dependence on petroleum fuels, mainly diesel, for the movement of vehicles, agriculture, and industries. A classic array of questions arises in this domain. For one, petroleum is a finite resource, and with increased world demand, supply

chains are under greater stress. Another is that the vaporization of fossil fuels remains one of the major sources of greenhouse gas emissions, increasing climate threats, and polluting the air while posing health threats to the masses (IEA, 2023). Thirdly, petroleum imports place many countries at economic and geopolitical instability. All of these issues combined compel one to identify and seek renewable, sustainable alternatives to conventional diesel-based fuels.

Biodiesel is a quick fix. Biodiesel is a sulfur-free, biodegradable fuel made from renewable feedstocks like vegetable oils, animal fats, or used cooking oils. It can be mixed with or used in place of petroleum diesel in existing engines with little or no modifications. When burned, biodiesel releases a lot less carbon monoxide, particulate matter, and lifecycle CO₂ than fossil fuels do. The US Department of Energy says biodiesel can reduce greenhouse gas emissions over its entire life cycle by up to 78% compared to petroleum diesel (DOE, 2024). Biodiesel production makes energy security stronger, adds to the fuel supply, and creates new opportunities to recycle waste in both rural and urban areas.

This project is not just about producing renewable fuel but also about valorizing waste. Using waste cooking oil (WCO) as the feedstock has economic and environmental benefits. WCO is cheaper than refined vegetable oils like soybean oil, which dominates global biodiesel production and doesn't divert food crops into fuel. Since feedstock cost accounts for 70-80% of biodiesel production cost, using WCO is a big economic gain (Atadashi et al., 2010). Moreover, converting WCO into biodiesel prevents improper disposal practices that can pollute waterways, clog sewage systems, and increase treatment costs for municipalities (Lam et al., 2010).

The problem we address has two parts: the environmental effects of fossil diesel use and the waste oils, which cause environmental damage. Through creating a biodiesel system using WCO,

this project develops an energy solution that is environmentally friendly while being economically sustainable.

Market Analysis

According to Grand View Research, Fortune Business Insights and Mordor, the global biodiesel market is projected to grow anywhere from 54 billion to 73 billion liters annually by 2030 (Biodiesel Market Size, Share & Trends Analysis Report, n.d., Biodiesel Market, 2025, Biodiesel Market Size & Share Analysis, n.d.) This growth can be attributed to emerging climate commitments, national biofuel mandates and the growing pressure to decarbonize the transportation industry. Biodiesel offers a sustainable, biodegradable and sulfur-free alternative to standard petroleum fuel sources, making it a more desirable fuel source in the global energy market.

Despite its environmental benefits, the widespread adoption of biodiesel is limited primarily by cost. Feedstock prices account for 70-95% of biodiesel production costs, with soybean oil and rapeseed oil representing the largest shares (Duti et al., 2016). Relying on refined oils not only inflates production costs but also raises concerns about the food vs. fuel conflict, as edible oils are diverted away from food supply chains. To overcome these challenges, increasing attention has been directed toward waste frying oil (WFO) and other high-free fatty acid (FFA) waste oils. These feedstocks are abundant byproducts of the food industry, cheaper than virgin vegetable oils and their conversion into biodiesel supports circular economy principles by reducing waste disposal problems (Duti et al., 2016).

However, the high FFA content complicates biodiesel synthesis, since in conventional base-catalyzed transesterification FFAs form soap when reacting with alkali catalysis, lowering yield and complicating purification. This makes a dual-step process like acid pretreatment followed by base-catalyzed transesterification both technically necessary and economically advantageous. The acid esterification step converts FFAs into esters, reducing soap formation, while the base step efficiently converts triglycerides to biodiesel. Together, these methods enable the use of low-cost waste oils at high yields while meeting international quality standards.

Europe, the United States, Brazil and Indonesia dominate global production but increasing policy support across Asia and South America is expected to further expand the sector. Waste frying oil represents one of the most attractive feedstocks for future biodiesel production. In the United States alone, approximately 100 million gallons of waste cooking oil are produced daily, while Europe generates 0.5-0.7 million gallons per day (Duti et al., 2016). Compared to soybean oil, which costs around \$751 per metric ton, WFO is 2-3 times cheaper, making it a compelling alternative for commercial-scale operations. The adoption of acid pretreatment plus base transesterification processes enables the cost-effective use of these high-FFA feedstocks, reducing overall manufacturing costs and improving competitiveness.

The biodiesel value chain for biodiesel production begins with the collection of waste frying oil from restaurants, households and food processing industries. The oil undergoes an acid pretreatment step where sulfuric acid and methanol are used to convert FFAs into esters. This step is followed by a base-catalyzed transesterification where triglycerides react with sodium hydroxide and methanol to produce fatty acid methyl esters (FAME) and glycerol. The product mixture then undergoes separation and purification through settling, water washing and drying to remove catalyst residues and water. Unreacted methanol is recovered through distillation and recycled and

the purified biodiesel is ultimately distributed as either a blend with conventional diesel or in pure form to meet ASTM D6751 and EN 14214 standards.

Biodiesel pricing has historically been higher than petroleum diesel which has hindered its widespread adoption. For example, in 2015 biodiesel from soybean oil cost \$3.55 per gallon compared to \$2.93 per gallon for petroleum diesel (Duti et al., 2016). Current biodiesel prices around \$4 per gallon depending on region (U.S. Department of Energy, 2024). Using WFO as a feedstock mitigates this cost disadvantage, since its lower price significantly reduces operating costs and improves profitability. The ability to integrate WFO into biodiesel production thus not only lowers costs but also creates value from a material that would otherwise require disposal.

Governmental policy and regulatory frameworks are another important driver of the biodiesel market. In the United States, the Renewable Fuel Standard (RFS2) mandates that biofuels, including biodiesel, be blended into the national fuel supply. In the European Union, the Renewable Energy Directive (RED II) enforces renewable fuel quotas in transportation. Additional incentives such as carbon credits and low-carbon fuel standards further improve the economic case for biodiesel, particularly when produced from waste oils with lower lifecycle emissions.

Adopting acid pretreatment followed by base-catalyzed transesterification therefore expands the usable feedstock base beyond refined oils to include cheap, abundant and sustainable waste oils. Studies have demonstrated that this dual-step pathway can achieve biodiesel yields of 77-98%, producing fuels that meet ASTM and EN standards (Duti et al., 2016). This process reduces production costs, minimizes waste disposal issues and strengthens the environmental case for biodiesel. It provides a technically robust and economically sustainable route to meeting future renewable energy demands. By combining waste valorization with established chemical

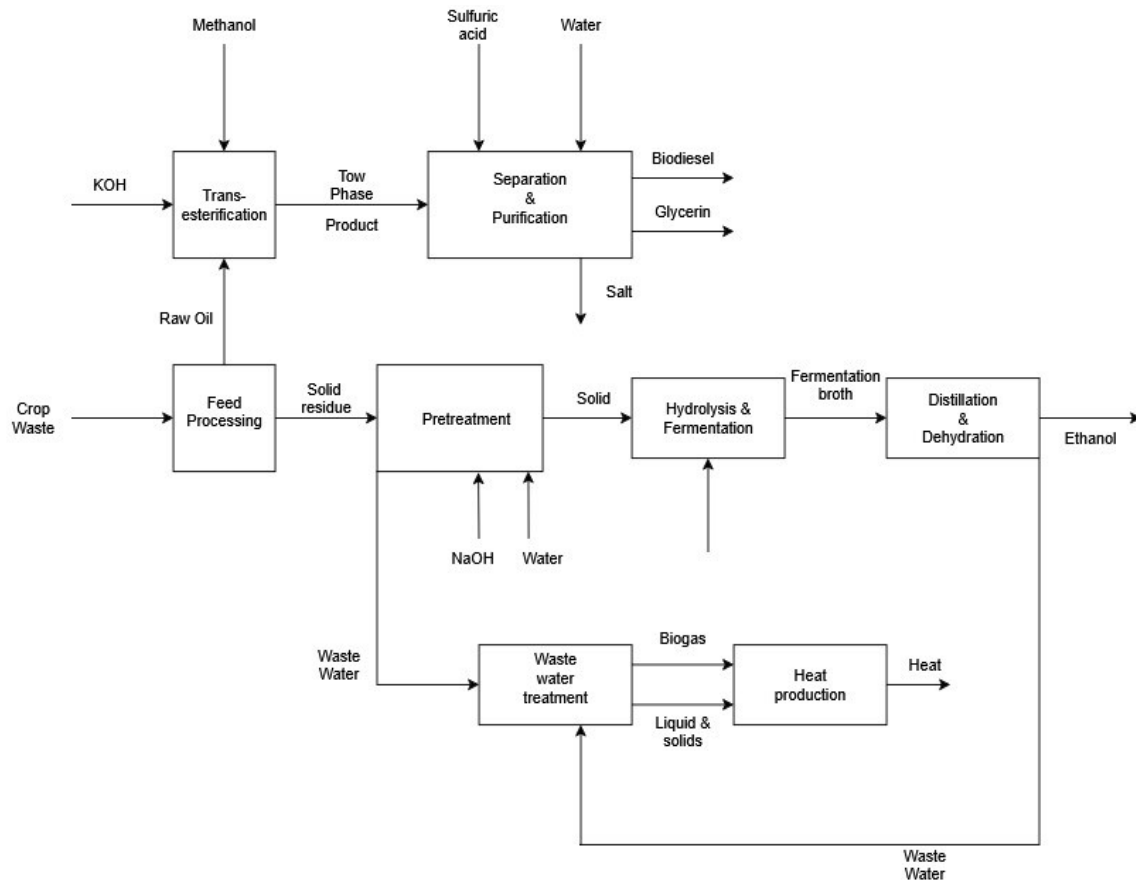
engineering practices, the dual-step process offers a scalable pathway toward reducing transportation sector emissions and enhancing energy security worldwide.

Process Description

Competing Processes

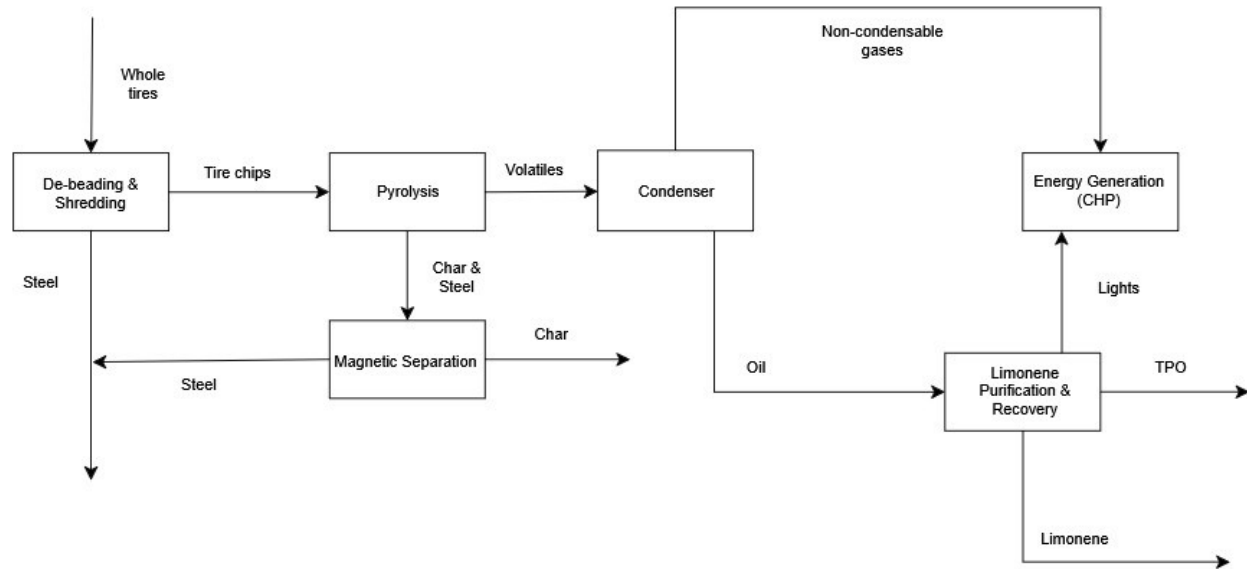
Creating fuel using crop waste

Crop waste is collected, cleaned, dried & shredded/milled (better surface area). The oil-bearing parts are then extracted (either mechanically or using solvents); another method would be to use an alcohol and catalyst (as seen). Under optimal/good conditions we often see a yield of 90-97% of the biodiesel. However, there are many factors that can play against this: particle size, moisture content, alcohol-oil molar ratio, catalyst type (as well as its concentration), and the reaction temperature (optimal:60-70C). The use of transesterification or other bio/thermochemical routes can yield fuels with reasonably good energy content (~37-41 MJ/kg) compared to the energy of petroleum diesel (~43-45 MJ/kg). Looking at the costs, we see that per gallon, we receive \$2.60 - \$4.66, which is heavily dependent on the plant size and conversion efficiency of the process.



Using rubber tires

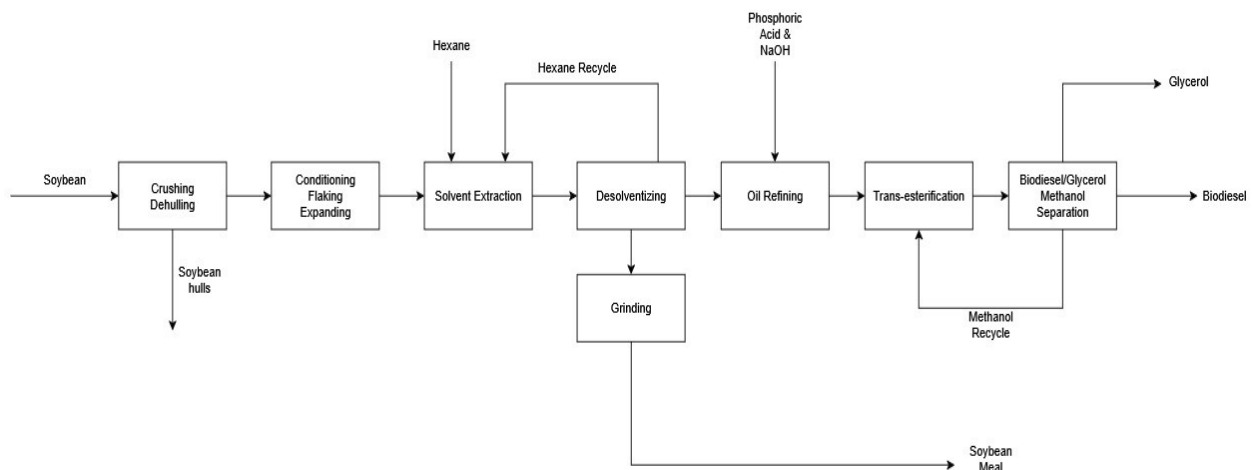
In optimized systems ~40-45 wt.% of the input tire material is converted into TPO (tire pyrolysis oil). The process itself is mainly sustainable as it produces non-condensable gases, which then can be used to heat the reactor, in turn lowering external energy use; along with this we also produce char/steel byproducts that can be sold. As sustainability provides lower greenhouse gas emissions than open burning/landfilling (the usual options for waste tires), we gain potential economic benefits, although the: energy use, transportation impacts and emissions of toxic compounds must be very closely monitored. Overall, the environmental footprint of this process is significantly smaller than that of conventional fossil fuel production/disposal of tires.



Using Soybean waste

As soybeans contain ~18-20% oil by weight (lower than any other oilseed), one bushel of soybeans produces ~1.5 gallons of biodiesel. The fuel produced by this has an energy content of ; ~37-40 MJ/kg, which is roughly 8-10% lower than normal petroleum diesel.

In terms of sustainability, the benefit of using soybeans is that there is already well-developed infrastructure (e.g. planting, harvesting, transporting and processing) that can help with production/material gain.

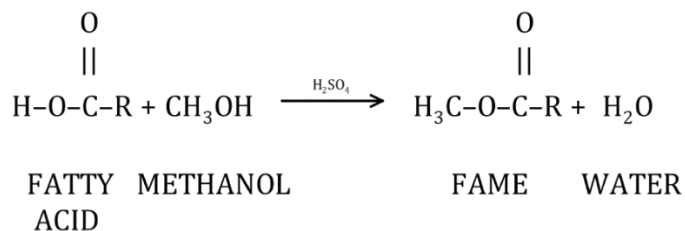


Chosen Process Detailed Description

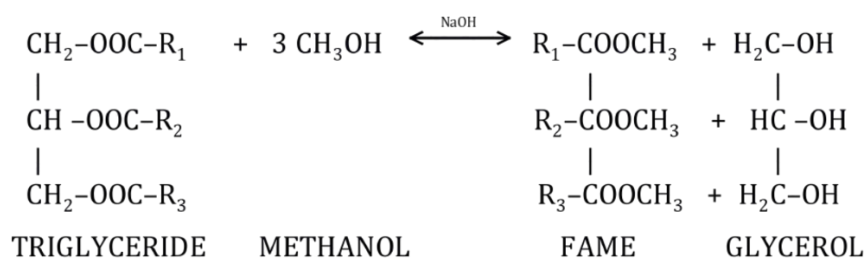
The selected production process for this research is a two-step method that consisting of acid pretreatment followed by base-catalyzed transesterification. Because waste frying oil (WFO) has a high concentration of free fatty acids (FFA), it needs to be treated before the transesterification reaction of triglycerides. During pretreatment, FFAs are converted to esters in the presence of an acid catalyst (usually H_2SO_4) to suppress soap formation and improve conversion (Alkhafaje et al., 2020).

The transesterification process involves a chemical reaction between the pretreated oil and an alcohol to produce alkyl esters in the presence of a base catalyst. Methanol is most commonly used because it is inexpensive and dissolves the catalyst efficiently (Dermibas, 2009 as cited in Duti et al., 2016). Among base catalysts, KOH or NaOH are frequently used for their high yields and faster reaction rates (Staravarache et al., 2006 as cited in Georgogianni et al., 2007). Although KOH produces a faster reaction, it is less effective overall, which is why NaOH is preferred for transesterification (Dias et al., as cited in Veljković et al., 2021).

In the process, WFO is first heated to 55°C in heater E-100 to lower viscosity. Then reacted with methanol and H_2SO_4 in the Pretreatment Vessel, an agitated tank reactor operated at 55°C and near atmospheric pressure. Here, free fatty acids (FFAs) react with methanol to produce fatty acid methyl esters (FAME), and water as shown in the reaction below:



The effluent from the pretreatment vessel enters the Methanol Separator, a distillation column that removes excess methanol, 95% of which is recovered and recycled into Mixer-1. Fresh methanol and NaOH are combined in Mixer-2 to form a sodium methoxide solution. This solution, long with the pretreated WFO is then sent to the Transesterification Vessel, an agitated continuous reactor also maintained at 55°C. The pretreated WFO is then sent to the Transesterification Vessel, where it reacts with Methanol and NaOH. Here, triglycerides in the oil react with sodium methoxide to yield FAME and glycerol according to the following reaction:



Excess methanol is again recovered and recycled into the Transesterification Vessel. The separated glycerol is removed from the process, and the FAME product is sent to a washing chamber to remove any residuals. Finally, the raw biodiesel is heated to about 100°C to be purified of any remaining water and then cooled in E-101. A process flow diagram is shown below:

Because high temperatures can have a reduction in conversion (Phan et al., 2008), the reactor vessels are maintained at around 55°C throughout the process to yield around 95% conversion. Also, 95% of methanol is recovered from both recycle streams.

Appendix

Design Basis

Target

- Annual production: 100,000 metric tons per year (100,000,000 kg/year)
- Operating schedule: 8,000 h/year
- Plant Location: Midwest
- Hourly production target: 12,500 kg/h FAME (biodiesel)

Feed & Spec.

- Feed: (WFO)
- Assumed composition: 90 wt.% triglycerides, 10 wt.% free fatty acids ($1,875 \frac{kg}{h}$),
- Product spec: ASTM D6751 / EN 14214 compliant FAME ($\geq 96.5\%$ purity)

Molecular Weights (assumed)

Fatty acid: $282 \frac{g}{mol}$

• Triglyceride: $885 \frac{g}{mol}$

• FAME: $296 \frac{g}{mol}$

• Glycerol: $92 \frac{g}{mol}$

• Methanol: $32 \frac{g}{mol}$

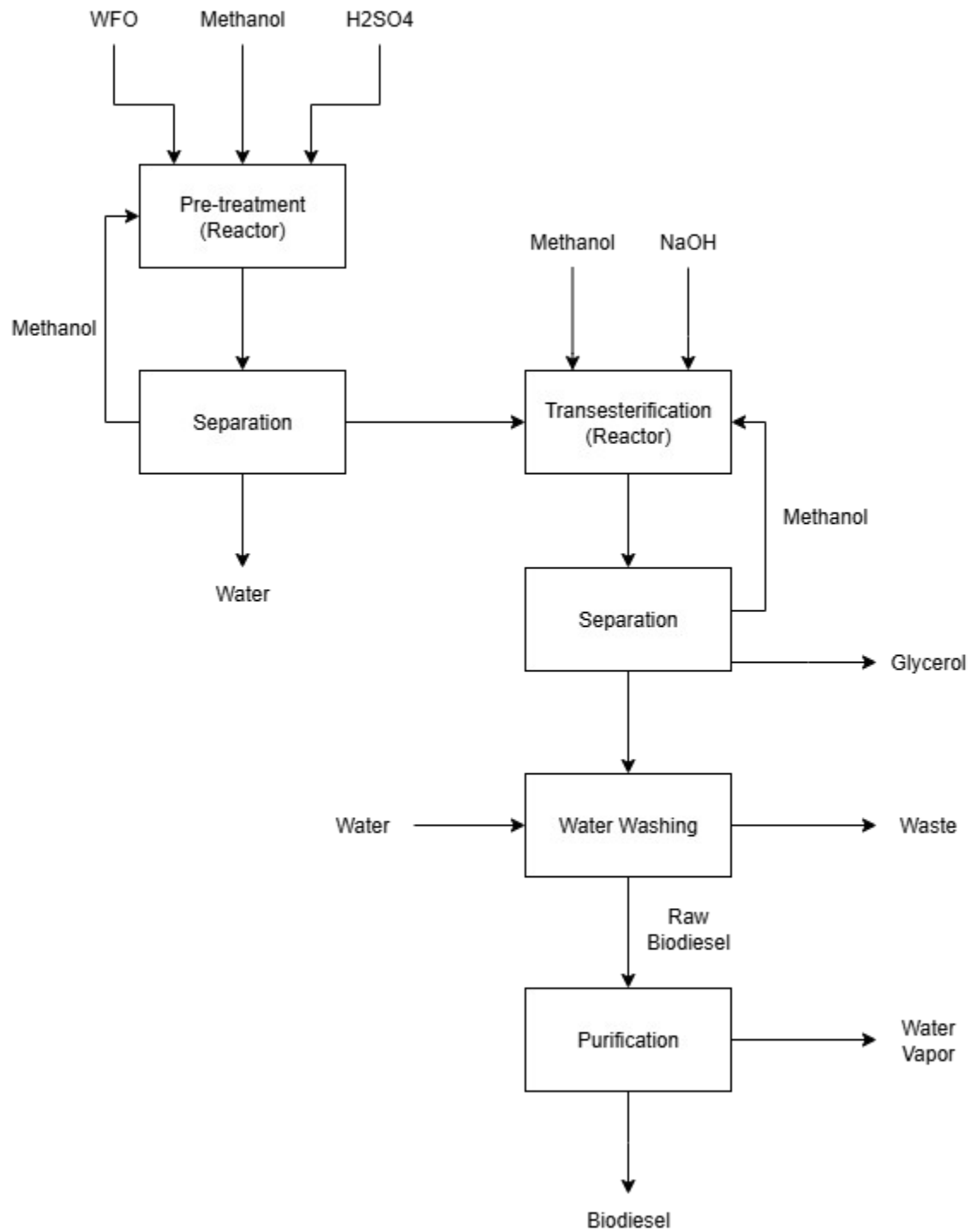
Targets of Operation

- Temperature: 55 °C
 - Pressure: Near-atmospheric
 - Residence time: ~60 min pretreatment, 60–90 min transesterification
 - Methanol recovery: 95% from both recycle loops

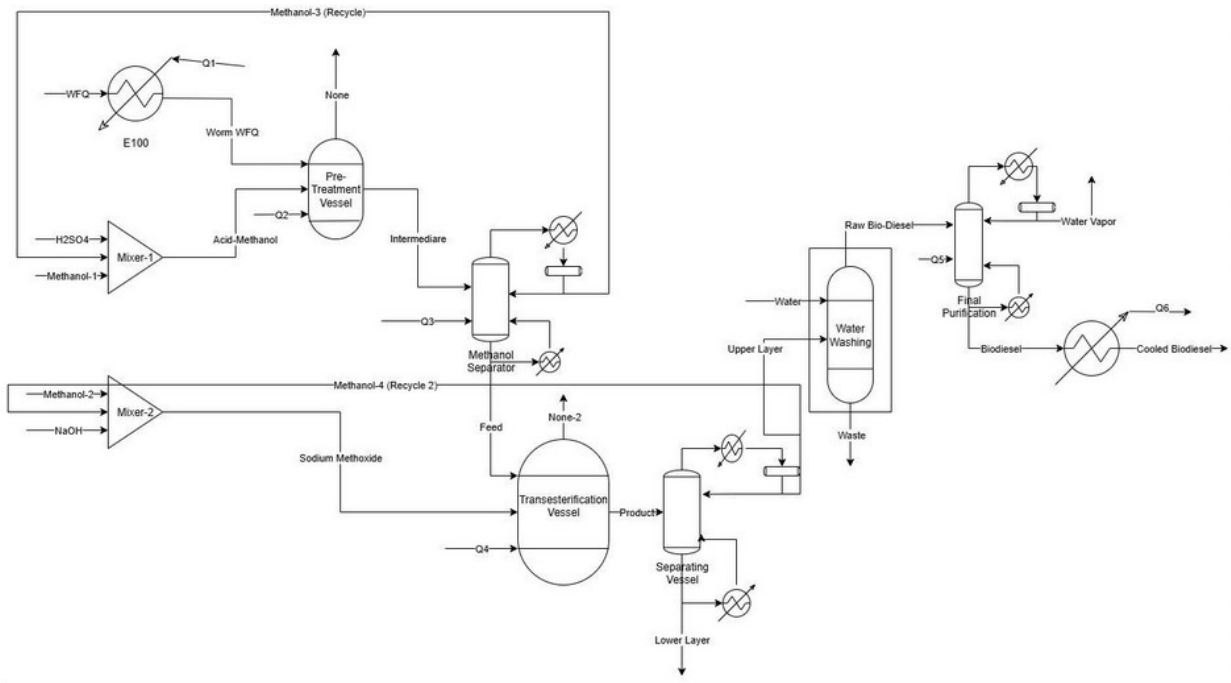
Stoichiometry & Flow

- FAME production
 - Methanol feed
 - Methanol stoichiometric consumption:
 - Fresh methanol makeup (after 95% recycle):
 - H₂SO₄ catalyst
 - NaOH catalyst
 - Glycerol by-product
 - Reaction water
- Wash water use

Block Flow Diagram (BFD)



Preliminary Process Flow Diagram



Rough Material Balance

-Target Product Biodiesel (FAME) = 100,000 metric tons/year

-Operation time: 330 days

$$\text{-Product rate} = \dot{m}_{FAME} = \frac{100,000 * 1000}{(300 * 24)} = 12,626 \frac{kg}{hr} = 27,836 \text{ lb/hr}$$

-Feed Stock (WCO): 10 wt.% FFA and 90 wt.% TG

-Reactor conversions: 95% FFA in acid pretreatment and 95% TG in the Transesterification reactor

-Methanol recovery: 95% recovery

-Molecular weights: FFA 282, TG 885, MeOH 32.04, FAME 296, H2O 18, Glycerol 92.09, all in

$$\frac{lb}{lbmol}$$

$$\text{- Total FAME Balance: } 0.95 * \left(\frac{0.1}{282} + \frac{3*0.90}{885} \right) m = 94.08, m = 29,070 \frac{lb}{hr}$$

- Feed of WCO:

$$n_{FFA, \in i} = \frac{0.10 * 29,070}{282} = 10.29 \frac{lbmol}{hr}, n_{TG, \in i} = \frac{0.90 * m}{885} = 29.57 \frac{lbmol}{hr}$$

-Extent of reactions:

$$-\xi_1 = 0.95 * 10.29 = 9.78 \frac{lbmol}{hr}, \xi_2 = 0.95 * 29.57 = 28.09 \frac{lbmol}{hr}$$

Stream	From → To	Component	Flow Rate (lbmol/h)	Flow Rate (lb/hr)
S1	WCO → Pretreatment	FFA	10.31	2907
S1	WCO → Pretreatment	TG	29.56	26163
S2	MeOH Feed 1 → Pretreatment	MeOH	206.17	6606
S3	Pretreatment → Water Separator	FAME	9.79	2899
S3	Pretreatment → Water Separator	H2O	9.79	176
S3	Pretreatment → Water Separator	FFA (unreacted)	0.52	145
S4	Water Separator → Waste	H2O	9.79	176
S5	Water Separator → Base Reactor	FAME (from pretreat)	9.79	2899
S5	Water Separator → Base Reactor	FFA (unreacted)	0.52	145
S5	Water Separator → Base Reactor	TG	29.56	26163
S6	MeOH Feed 2 → Base Reactor	MeOH	177.38	5683
S7	Base Reactor → Phase Separation	FAME (new)	84.25	24939
S7	Base Reactor → Phase Separation	Glycerol	28.08	2586
S7	Base Reactor → Phase Separation	TG (unreacted)	1.48	1308
S7	Base Reactor → Phase Separation	FFA (carried)	0.52	145
S8	Phase Separation → MeOH Recovery	MeOH	383.55	12289
S9	MeOH Recovery → MeOH Header	MeOH (recycle)	364.37	11674
FRESH	Fresh MeOH → MeOH Feed Header	MeOH (makeup)	113.22	3628
S10	MeOH Recovery → Loss/Purge	MeOH (loss)	19.18	614
S11	Wash/Dry → Biodiesel Product	FAME (product)	94.04	27836
S12	Phase Separation → Glycerol Processing	Glycerol	28.08	2586

A steady-state material balance was performed on a product basis of 100,000 t/yr biodiesel ($27,836 \frac{lb}{hr}$). Using 95% conversion of FFAs in pretreatment and 95% conversion of triglycerides in base transesterification, the required WCO feed is $29,070 \frac{lb}{hr}$ (10 wt.% FFA, 90 wt.% TG). Pretreatment generates $9.78 \frac{lbmol}{hr}$ FAME and $9.78 \frac{lbmol}{hr}$ water; the water is removed prior to the base transesterification step. Transesterification generates $84.27 \frac{lbmol}{hr}$ FAME and $28.09 \frac{lbmol}{hr}$ glycerol. The final biodiesel product rate is $94.08 \frac{lbmol}{hr}$ ($27,836 \frac{lb}{hr}$), which meets the target capacity. Methanol is sent to the reactors at a rate of about $384 \frac{lbmol}{hr}$, and 95% of it is recovered through distillation (about $364 \frac{lbmol}{hr}$). The remaining 5% is lost, so the amount of fresh methanol needed to make up for this loss is about $113 \frac{lbmol}{hr}$ (about $3,630 \frac{lb}{hr}$). Glycerol ($28.09 \frac{lbmol}{hr}$) and water ($9.78 \frac{lbmol}{hr}$) are by-products. A small amount of unreacted oil slip (about $1,450 \frac{lb}{hr}$ TG+FFA) is polished to meet ASTM fuel standards.

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